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PDA
Parenteral Drug Association

Technical Report No. 13-2

Fundamentals of an Environmental Monitoring Program
Annex 1: Environmental Monitoring of Facilities Manufacturing Low Bioburden Products

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Fundamentals of an Environmental Monitoring Program Annex 1: Environmental Monitoring of Facilities Manufacturing Low Bioburden Products

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1.0 Introduction

Environmental monitoring (EM) is used to evaluate the effectiveness of contamination controls used in the manufacture of pharmaceutical products. EM programs need to be designed and implemented based on the needs of the specific facility and processes involved. PDA *Technical Report No. 13 (Revised 2014): Fundamentals of an Environmental Monitoring Program* (TR 13) focuses on the environmental requirements for sterile product manufacturing ***(1)***. This addendum will provide elements to consider when designing a risk-based EM program to support the manufacture of low bioburden products using low bioburden processes.

Regulations and guidance are readily available for EM requirements supporting sterile, nonsterile, and advanced therapy medicinal products (ATMP) manufacturing processes ***(2–6)***. Guidance for EM supporting the manufacture of low bioburden products is very limited. Where it is, the documents often cite resources intended for aseptic processes and then recommend establishing a risk-based approach to designing an EM program to support low bioburden processes ***(7, 8)***. This is largely due to the fact that most manufacturing operations where low bioburden control is required combine elements of both sterile and nonsterile processing. However, no specific guidance identifies the elements to which these sterile and nonsterile guidelines should be applied.

This document provides a means of determining the critical aspects, assessing the associated risks, and designing the appropriate EM program for areas where bioburden control of manufacturing processes is required. It can be applied across many types of manufacturing operations used to produce pharmaceutical products including biologics, small molecules, ATMPs, and others. This document should be viewed as technical guidance and is not intended to establish any voluntary or mandatory standards.

1.1 Purpose

This addendum complements the existing PDA TR-13 and is not meant to change or modify its content. The addendum provides recommendations for the development of an environmental control and monitoring program for classified areas and the associated GMP processes used for manufacturing operations requiring bioburden control in the biotechnology and pharmaceutical industries. The content includes a review of regulatory requirements and the development of risk assessments based on the criticality and complexity of processes, as well as industry examples of these risk-based approaches.

Risk assessment tools such as Failure Mode Effects Analysis (FMEA), Hazard Analysis and Critical Control Points (HACCP), or hybrid approaches can be used for these evaluations. This addendum presents a hybrid approach that is intended to help analyze operational risk in order to develop appropriate levels of EM required to monitor operations. When implementing these tools in the design of any EM program, the resulting program should fit the intended purpose. It should not unnecessarily impede the operation or introduce additional risk to the process.

1.2 Scope

TR-13-2 focuses on environmental controls and monitoring programs for classified areas where manufacturing processes are required to ensure a low bioburden level of the material being produced. Although a comprehensive EM program addresses both viable and nonviable aspects of the facility, this addendum concentrates primarily on the viable aspects as these parameters have a greater potential to impact product and process quality related to bioburden. This document is not intended to address EM programs for traditional aseptic or sterile manufacturing, which are covered in TR-13, nor does it address traditional nonsterile dosage forms.

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2.0 Glossary

The terms and definitions contained in this glossary are specific to the content of this technical report. Although some of these definitions will match those terms as defined in PDA TR-13 and/or other PDA technical reports (noted for each definition), others have been intentionally altered or developed to address the specific content and intent of this addendum and the attributes unique to facilities used to manufacture low bioburden products.

Adverse Trend
EM data (including microbial and particulate) that is consistently increasing in one direction. Adverse trends are system- or area-specific, should be evaluated on a case-by-case basis, and should consider factors such as contamination recovery rate, excursion rate, and individual results.

Airborne Particulate Count (Total Particulate Count)
The total number of particles of a specific size per unit volume of air ***(1)***.

Airborne Viable Particulate Count (Total Airborne Microbial Count)
The recovered number of colony-forming units per unit volume of air ***(1)***.

Bioburden
The total number of microorganisms per unit of material prior to sterilization ***(1)***.

Bioburden Reduction Step
A process utilizing a method that has been demonstrated to lower the bioburden level in a given sample.

Processing Area
The area in which processing occurs, with the appropriate environmental (e.g., gowning, cleaning/disinfection, training) and engineering (e.g., HVAC, facility design, material/personnel flows, differential pressures) controls in place to reduce the risk of microbial or particulate contamination.

Classified Area
An area typically classified according to ISO standards.

Cleaning
Chemical or physical means used to remove soil and/or microorganisms from surfaces ***(1)***.

Cleanroom
A room designed, maintained, and controlled to prevent particle and microbiological contamination of a drug product. A cleanroom is assigned and reproducibly meets an appropriate air cleanliness classification ***(1)***.

Critical Operations Monitoring
Monitoring performed throughout the duration of a process using methods designed to monitor the environmental quality of the process and to capture all interventions, transient events, and any system deterioration. The design and degree of the monitoring performed should be based on the risk to the process or product being monitored (e.g., closed vs. open systems, use of biological safety cabinet, unidirectional flow hood, or other containment systems).

Disinfection
The chemical or physical inactivation of a bioburden on inanimate surfaces. Typically, this requires a minimum three-log (3-log) reduction of vegetative microorganisms and a two-log (2-log) reduction of bacterial spores to be achieved in validation ***(9)***.

Environmental Control Parameters
Conditions and corresponding measurements as associated with facilities and equipment used in the control of a manufacturing area that may impact the identity, strength, quality, or purity of a product. Among such parameters are airflow rates and patterns, pressure differentials, materials and personnel flow, temperature, and relative humidity, as well as nonviable and viable particulates ***(1)***.

Environmental Monitoring

Dynamic (Operational) Monitoring
The condition where production operations are actively occurring as defined by the individual firm. Personnel may or may not be present at this time.

Routine Monitoring – Frequency-Based/Predetermined/Nonbatch-related
Monitoring of a facility at a predetermined frequency, as defined by an approved procedure, to provide an overall assessment of the state of control of the facility.

Batch-related – In-Process or During-Process Monitoring
Monitoring performed during predetermined steps of the production process, as defined by

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an approved batch record and/or approved procedures or protocols. This can be performed either in the presence or absence of the product.

Continuous

Batch-related monitoring, as defined by an approved batch record, to ensure that a state of control is maintained during processing and that any aberrant events are detected. Frequency of monitoring is determined by a risk assessment.

Closed System

A process system with equipment designed and operated such that the product is not exposed to the room environment at any point in the process. For most biological operations, the system or fluid pathway must remain free of extraneous contamination. Examples include, but are not limited to, a cell culture bioreactor or cell therapy bag assembly.

Functionally Closed System

A process system that may be routinely opened (e.g., to install a filter or make a connection), but is returned to a closed state during processing. In biological operations, for example, a functionally closed system is often employed when a low level of extraneous contamination may be permissible (such as purification skids used in cell-culture operations).

Open System

A process system that exposes the product to the room environment. In this type of system, the room environment is controlled to minimize the risk of extraneous product contamination. Examples include, but are not limited to, processing steps, such as media preparation or supplementation, or inoculum preparation.

Low Bioburden Product

Product manufactured within a controlled and monitored environment in which the final drug product or process intermediate, as applicable, requires bioburden control, but is not required to be sterile (e.g., biological drug substance produced by mammalian cell culture).

Nonviable

A term used in reference to particulates that are not capable of living, growing, or developing and functioning successfully ("unable to divide" or "not capable of reproducing") ***(1)***.

Process Control Parameters

Conditions and corresponding measurements associated with the manufacturing process that may affect the identity, strength, quality, potency, and purity of a product. Examples of parameters of concern include bioburden, process rate, weight, volume, temperature, and pressure ***(1)***.

Risk Analysis

The estimation of the risk associated with the identified hazards ***(1)***. The following are commonly used tools that can be used to perform risk analysis ***(10, 11)***.

- FMEA – Failure Mode Effects Analysis
- HACCP – Hazard Analysis and Critical Control Points

Sanitization

Reduction of microbial contaminants to safe levels as judged by public health requirements for the specific country ***(1)***.

Sterilization

Validated process used to render product free from viable microorganisms ***(1)***.

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3.0 Low Bioburden Products and Processes versus Traditional Sterile and Nonsterile Products and Processes

Sterile drug manufacturing is designed to produce a final sterile product through sterilizing process steps and subsequent manufacturing processes and controls. In contrast, manufacturing processes that require bioburden control are designed to limit the introduction of microorganisms into the process.

Based on the physical properties of the materials being manufactured or, in some cases, regulatory requirements, mitigative steps designed as bioburden control mechanisms, such as the use of antibiotics, preservatives, solvents, low or high pH, sterile filtration and heat, may not be possible. In many cases, the process streams for these processes support the proliferation of any microorganisms present, often requiring control mechanisms during manufacturing operations to prevent or reduce microbial growth. Specific process steps taken if microorganisms are present, such as purification and ultrafiltration, often reduce bioburden levels although, in general, they are not specifically designed to do so. Due to the inherent susceptibility of the manufacturing process in these cases, an added focus must be placed on minimizing the ingress and proliferation of microorganisms and their metabolites that can affect product quality, safety, and stability.

Control of the process and processing environment is a major component of reducing the potential for product stream bioburden. Elements of control include the design of the facilities, utilities, and processing equipment (closed/open), as well as personnel and procedural measures. EM is used as a tool to measure the efficacy of these controls in a manufacturing facility.

The use of a local unidirectional airflow hood (UDF) for specific open manufacturing steps is a common practice where bioburden control is desired. These units are designed to produce a localized EU Grade A/ISO 5 environment. In this application, the units are not being used to provide protection for a sterile manufacturing-related operation but rather a nonsterile manufacturing operation as part of a bioburden control program. In such cases, based on their use and a thorough risk assessment, the placement of the UDF in Grade C environments is often acceptable. The level of gowning associated with their use should also be evaluated and requirements set based on the operation being performed and the level of control required.

The implementation and utilization of a quality risk management program provides scientific rationale for the manufacturing controls and EM program developed for these types of manufacturing processes ***(10, 12)***. Quality risk management is predicated on understanding the specific risk factors of the environment and the process used to manufacture the product.

Once the manufacturing process and the potential contributors to the level of bioburden (e.g., environmental factors, facility and/or equipment design) have been identified and characterized, a thorough and well-documented risk assessment should be conducted to identify the points of potential microbial ingress. The manufacturer may incorporate specific components of a regulation or guidance document into the risk assessment as it applies to a particular process or step; this addition may vary depending on the intent of the risk assessment. Upon completion of the risk assessment, supporting programs can be developed to mitigate the risks identified. While all risks may need to be monitored for a period of time, any risk that cannot be mitigated should continue to be monitored.

When a manufacturing area has been classified in line with a regulatory or industry classification scheme, the EM program is often expected to meet the applicable requirements regardless of how the area is used. For example, if an area is classified as EU Grade A, the EM program and other supporting programs (e.g., gowning, cleaning, and sanitization) must reflect the requirements contained in the corresponding regulatory document. In many cases, these guidance documents are designed specifically to address the manufacture of sterile medicinal products, not nonsterile manufacturing operations designed to control but not eliminate the bioburden load present.

Consider the scenario of an aseptic transfer of materials in the UDF to minimize the introduction of additional bioburden. This is a nonsterile manufacturing step. A risk assessment may demonstrate

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that only certain aspects of the EU Grade A classification requirements are pertinent to the process. Monitoring performed during the manufacturing process should be aligned with the risk assessment and reflect the monitoring needed to support the process. For example, in a situation where the concern is focused on the addition of bioburden (microorganisms) to the manufacturing operation the risk assessment may conclude that the need to perform total particulate monitoring during the manufacturing operation should be reduced or eliminated. In that case, the total particulate monitoring would still be performed at a set frequency to ensure the UDF is continuing to operate appropriately but not during the actual manufacturing step. Similarly, the surrounding room environment should be determined based on the assessment of the manufacturing process and the control needed. For traditional nonsterile operations, for example, the EU Grade C requirements may be more appropriate based on the operation being performed. An alternative could be to classify the area according to ISO 14644 (e.g., ISO 5, 7, 8), as appropriate for the specific process and area ***(13)***. This approach must be documented in the site risk assessment. The supporting EM and other control programs should be designed to monitor the room or facility to the levels needed for the specific processes being performed.

An additional example where the use of a risk assessment may be of value is in the use of a UDF where preparation of the inoculum used for seeding of a bioreactor is being performed. In this example, during mammalian cell culture preparation, a designated room is used to prepare the inoculum for seeding the bioreactor. In the scale-up process, aseptic techniques are used to transfer the contents of a thawed cryovial to a sterile T-flask (or similar container). This activity is performed in a UDF to protect the process from the introduction of microorganisms, as microbial contamination would be the largest concern. The UDF is classified as EU Grade A because an aseptic technique must be performed in order to exclude the introduction of extraneous contaminants. This process takes less than two minutes to execute. Based on the air quality criteria described in *EudraLex, Volume 4, Annex 1: Manufacture of Sterile Medicinal Products*, both continuous nonviable particulate counts and active airborne viable monitoring would be required during the manufacturing process based on the UDF classification as Grade A ***(2)***. This would present some challenges if applied. Annex 1 has generally been interpreted to expect a sample size of one cubic meter of air for both types of monitoring, which takes between 5 and 35 minutes to collect depending on the type of equipment used. Considering that the scale-up process takes only two minutes to execute, the requirements for the sample volumes would be difficult to satisfy; however, if the process only takes two minutes, the monitoring would encompass the entire process and meet the intent of the regulation. In this case, a risk assessment should be completed to evaluate the process and monitoring requirements. If the primary concern is the introduction of viable organisms, the risk assessment may conclude that performing in-process total particulate counts not only provides little value but increases the risk to the operation based on the distraction it creates for the operator (assuming the use of manually operated particulate counters) and should not be conducted during the process while viable monitoring is of greater importance. As presented in the aseptic transfer example above, total particulate monitoring would be expected at an established frequency to ensure the biological safety cabinet is performing properly.

A number of well-known risk assessment tools are available, such as flow charts, check sheets, FMEA, and HACCP. In order to perform a successful risk assessment, the following must be included to ensure that all aspects of the process are appropriately addressed:

- Knowledgeable representation from the functional areas involved in the process (e.g., validation, manufacturing, quality, facilities, microbiology)
- Defined risk questions
- Detailed process map
- Facility map, including personnel and material movements
- Production batch records and standard operating procedures

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4.0 Environmental Risk Assessment

Several risk assessments need to be performed as part of an overall facility risk assessment; these can include a facility (also called an environmental) risk assessment, a process risk assessment, and an EM risk assessment. The environment (or facility) risk assessment should determine the suitability of the facility or environment in which the process is conducted. A process risk assessment should include identification of potential sources of contamination and associated inherent risks to the process from an operational perspective, including points of potential contamination. The EM risk assessment determines how the contamination risks should be addressed within an effective EM control program. All are part of the overall contamination control strategy for a facility.

This technical report focuses on the EM risk assessment that may be performed to:

- Develop or expand programs used to identify, control, and/or monitor contamination risks to determine:
  - Appropriate level of facility control for the process being performed
  - EM locations, sampling frequency, and sampling methodology
  - Appropriate environmental controls to be employed in manufacturing areas based on the type of processing performed
  - Develop or revise area and material transfer or cleaning and/or disinfection practices
- Determine gowning strategy and controls required to prevent contamination by personnel
- Determine appropriate personnel, material, and waste flows
- Develop or expand a bioburden and endotoxin-burden monitoring program

EM risk assessments can also be used to investigate a contamination event or trend, such as:

- Microbiological contamination in a process train or critical utility system
- Adverse data trend in a microbial measurement system (i.e., bioburden/endotoxin, EM, or critical utility system monitoring)
- Identification and evaluation of a potential source of contamination or atypical recovery

4.1 General Guidelines

Risk assessments highlight manufacturing process steps or room locations that exhibit the greatest opportunity for introduction of potential contamination to the product or process stream, a critical utility system, or the environment. A process flow map based on production records, applicable standard operating procedures (SOPs), personnel/equipment/material flow diagrams, and piping and instrumentation diagrams is used to identify individual process steps. Using a process flow map helps ensure that each risk assessment is performed systematically and that all key aspects of the operations or systems are considered. As a best practice, the system-owner maps out the process or systems prior to development of the risk assessment by the multifunctional team to ensure that the scope of the risk assessment is clearly identified.

Tours of the facility and manufacturing operations using the process flow map are also recommended to collect information and determine the criticality factors relevant to the risk assessment being performed (see **Table 4.3.1-1** and **Table 4.3.1-2**). Only the criticality factors that apply to the scope of the risk assessment should be selected.

4.2 Criticality Factors

Criticality factors assist in identifying contamination hazards by determining how each factor might impact the potential for product, process, or environmental contamination. The criticality factors are based on the various sources of contamination.

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Eight major potential sources of adventitious microbial contamination sources and their relevance to the risk assessment are discussed in **Sections 4.2.1–4.2.3.5**, topics that should be considered applicable to any facility and addressed through the risk assessment.

4.2.1 Personnel

The human body is the source of greatest risk of contamination in the manufacturing environment. The design of the EM program should consider the risk to the various environments based on the engineering and procedural controls designed to minimize the impact of personnel on the environment. Some potential controls are:

- Establishing maximum personnel load for the room or area
- Facility designs that eliminate the need for entrance into manufacturing areas (e.g., viewing galleries)
- Use of facility-dedicated shoes and clothing
- Gowning and/or personal protective equipment requirements and glove/sleeve sanitization procedures, as dictated by the specific process or process step
- Unidirectional vs. bidirectional personnel flows
- Use of good aseptic working practices
- Personnel training and qualification

4.2.2 Materials/Waste

The second major source of potential contamination is the transfer of materials into and waste out of the manufacturing and support areas and, potentially, into exposed product. All ingredients, processing aids, filters, components, labels, packaging materials, etc., must be disinfected (or sterilized if determined appropriate) and managed in a manner that reduces the risk of transferring contamination to the manufacturing and support areas. Waste should be managed in a manner that ensures a low risk of microbial and/or product cross-contamination. A comprehensive EM program must include sampling locations and frequencies to provide timely indications that materials and waste are being transferred and handled appropriately.

4.2.3 Facility Design and Construction

The design of the manufacturing and support areas and the implementation of the following minimum requirements are needed to facilitate cleaning and disinfection activities, as well as minimize the potential for contamination of those areas and the product (this list is not exhaustive):

- All surfaces should be smooth, impervious, nonabsorbent, unbroken, and compatible with appropriate agents
- The amount of exposed surface area should be minimized to the extent possible (e.g., pipes should be contained in interstitial spaces wherever possible)
- Construction materials should resist chipping, flaking, oxidizing, or other deterioration
- Wall and floor junctions should be curved, rather than meeting at sharp angles
- All flooring materials should be seamless
- Any recesses should be reachable and cleanable
- Ledges, shelves, cupboards, and equipment where particles or microbes can settle and accumulate should be minimized

The facility design and construction should be appropriate for the proposed manufacturing process and product. If during the execution of the risk assessment, the facility is found to be inappropriate in any way, the deficiency should be addressed through appropriate mitigation and the efficacy verified through the EM program.

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4.2.3.1 Facility Cleaning and Disinfection

The types of cleaning and disinfection agents (i.e., sanitizers, disinfectants, and sporicides), methodology, and frequency should be considered when designing an EM program, and the frequency of cleaning and disinfection activities should be justified based on historical EM data for existing facilities. For new facilities, this frequency should be determined through appropriate in-situ qualification as defined in PDA *Technical Report No. 70: Fundamentals of Cleaning and Disinfection Programs for Aseptic Manufacturing Facilities* ***(4)***.

4.2.3.2 Process Design

The design of the process should be considered when establishing the EM program. The degree of manual manipulations, number and complexity of required steps, and the process duration all affect the potential for microbial contamination to enter the system. Therefore, each step of the process should be considered individually in the risk assessment. For example, open operations may provide more opportunity for product contamination, whereas closed or hard-piped systems and isolators can significantly reduce the risk of contamination as they prevent exposure of the process stream to the environment and personnel. From an operational standpoint, additional bioburden reduction steps, such as filtration, should be considered.

4.2.3.3 Nature of the Product

The nature of in-process intermediates and final product should also be considered, such as the potential for growth-promoting characteristics. For example, upstream cell culture media would be highly sensitive to microbial proliferation whereas potential downstream purification materials with high salt buffers may prohibit microbial proliferation.

4.2.3.4 Environmental Air

Controlling air as a source of contamination is an important factor to consider when designing an EM program. Points to consider include, but are not limited to:

- Heating, ventilation, and air conditioning design
- Number of air changes per hour
- Use of localized filtered air (e.g., UDFs)
- Use of pressurized airflow cascades
- Use of transitional spaces or rooms (e.g., airlocks and gowning rooms)
- Results of airflow visualization studies (e.g., smoke studies)

4.2.3.5 Utilities

The main types of utilities used in controlled environments are water, compressed air or gases, vacuums, and drains. Risk of potential contamination from exposure to these utilities should be considered when determining sampling locations and frequencies for an EM program. How these utilities impact the process and product should also be considered, as they impact the quality and design of the resulting EM program.

4.3 Risk Assessment Design

Many methods are available for designing and implementing a risk assessment. The selection of the method should be based on the process in question and the relevant criticality factors selected. For the purposes of illustration, **Sections 4.4** and **4.5** present examples of hypothetical risk assessment designs for a typical mammalian cell culture process (upstream).

Each risk-ranking is performed in accordance with the risk tool selected. The risk control activities are performed according to the criteria specified in the applicable risk tool methodology. If a location

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or process step is not well controlled and/or monitored and considered a high risk, then that location or process step automatically becomes a potential source of microbial ingress, and tightening contamination controls should be considered.

4.3.1 Criticality Factors

Criticality factors are criteria that are divided into three main groups: process step, potential contributors to contamination, and unusual/unexpected sources of contamination. **Tables 4.3.1-1** and **4.3.1-2** provide examples of elements that aid in determining the level of risk—High, Medium, or Low—and the criticality factors of each. These elements may change based on the manufacturing operations, equipment, facility design, or other components. The criteria may be numbers or values but need to be able to allow the user to evaluate the risk on a scaled basis. The criticality and weighting for each criterion may vary depending on the area and process or other factors.

4.3.1.1 Group 1 – Process Step

Each stage of the manufacturing process is considered a process step. The process steps shown in **Tables 4.3.1-1** and **4.3.1-2** are examples of those that would be used by the risk assessment team. These may differ based on the products being manufactured, manufacturing process, stage of the process, equipment set-up, facility age, and controls, among other elements.

4.3.1.2 Group 2 – Potential Contributors to Contamination

These criticality factors include the manufacturing process, environmental factors, facility or equipment design, and contamination control programs implemented, any of which may contribute to the potential for contamination. For the factors in this group, enhanced contamination controls may be identified during risk mitigation. In this mammalian cell culture example, the potential sources of contamination are shown in **Tables 4.3.1-1** and **4.3.1-2**.

4.3.1.3 Group 3 – Unusual/Unexpected Sources of Contamination

Group 3 evaluates sources of contamination that are extraneous to the design intent of the environment, equipment, or process. Any unusual or unexpected source of contamination is considered a high risk that must be mitigated. This may include faulty or leaking equipment or utility connections or breaches in the integrity of the manufacturing environment that are detected during a facility walk-through. Since the source of this hazard is unexpected and must be mitigated, controls should be implemented, and it should be monitored until the unexpected source of contamination has been eliminated. Examples include, but are not limited to, a reverse in pressure differentials due to a power failure, presence of contractors and their equipment, damage to the facility surfaces, or condensation on piping.

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Table 4.3.1-1 Examples of Criticality Factors for the Immediate Process Area for Mammalian Cell Culture

Process Step	High Risk	Medium Risk	Low Risk
Group 1: Process Step			
Manufacturing Process Step	Critical manufacturing steps • Final formulation • Seed inoculation • Cell-bank preparation • Vial-thawing and cell-splitting • Bead addition steps • Vector transduction	• Kitting of parts to be sterilized • Media filtration • Spiking • Post-wash filtration	• Aliquoting of samples • Equipment cool down • Equipment assemblies • Tubing welds
Group 2: Potential Contributors to Contamination			
Open, Functionally closed or Fully Closed System	Product exposed directly to the environment; open system connections	Open system connections with local protection (i.e., LFH/UDF)	• Functionally or Fully Closed system (e.g., pressurized or integrity controlled) • Use of established closed connectors
Preparation of Equipment/Materials	Equipment only cleaned and sanitized	Some equipment cleaned and sanitized; some parts cleaned and sterilized	All equipment cleaned and sterilized
Complexity of Equipment Train/Process	Manual assembly or manual manipulations; high number of inputs into the system (manual or automated)	Semi-automated with some manual manipulation; medium number of inputs into the system	Completely automated system; low number of inputs into the system
Length of Time Required to Perform the Critical Process Step/Operator Intervention	4–24 hours of processing	< 4 hours of processing	< 30 minutes of processing
Length of Time Operator is Required to be Physically Present in the Immediate ISO 5/Grade A Area	4–24 hours	< 4 hours	< 30 minutes
Classification of the Surrounding Area	ISO 9/Grade D or ISO 8/Grade C	ISO 8/Grade C	ISO 7/Grade B
Surrounding Room Temperature (Room outside of the LFH/UDF)	Warm (>25 °C)	Room Temperature (20-25 °C)	Cool (<20 °C)

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(Table 4.3.1-1 Continued)

Process Step	High Risk	Medium Risk	Low Risk
Level of Activity in Surrounding Room	Common batch-processing area (more than 1 operation occurring in room)	• Support area processing • Material/equipment prep area • Storage of sterilized parts and equipment (if dedicated to the same process)	• Dedicated processing area • Storage of closed system vessels
Number of Personnel in Surrounding Area	Maximum validated personnel load	More than one operator present, but not at the maximum validated load	Only one working operator present
Personnel Gowning	Operators gowned in street clothes with a dedicated, nonsterile lab coat, sterile sleeves, and sterile gloves (or similar)	Operators gowned in scrubs or single-use lab coats, sterile sleeves, and sterile gloves (or similar)	Operators fully gowned in sterile gowns, masks, sleeves, and gloves
Disinfection of the ISO 5/Grade A Area	Performed at a scheduled frequency (e.g., weekly or daily) regardless of the level of activity)	Performed following each operation	Performed prior to and following each operation
Frequency of Material Transfer into the ISO 5/Grade A Area	Multiple entrances and exits of materials into/out of the area throughout the duration of the process (e.g., filling of multiple containers)	Minimal movement of materials into/out of the area throughout the duration of the process (e.g., a single exchange)	All items placed into the area prior to start of work and no movement into/out of the area during process
Disinfection of Manufacturing Materials to be Transferred into the Environment	Sanitized with 70% IPA	Sanitized with a disinfectant and/or sporicidal agent	All required materials sterilized and multi-bagged
Group 3: Unusual/Unexpected Sources of Contamination			
Unusual/Unexpected Sources of Contamination	Unexpected contamination detected in an area, such as: • Leaking equipment or utility connection • Breach in integrity of environment or unidentified leakage of other equipment or materials (e.g. bags, connections) • Damage to facility surfaces	N/A	N/A

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Table 4.3.1-2 Examples of Criticality Factors for Supporting Area for Mammalian Cell Culture

Criticality Factors	High Risk	Medium Risk	Low Risk
Group 1: Process Step			
Manufacturing Process Step	Critical manufacturing steps • Final formulation • Seed inoculation • Cell bank prep • LFH/UDF	Downstream operations • Purification • Sterilizer unload • Kitting of parts and materials to be sterilized	Upstream operations • Cell culture • Media/buffer preparation • Support areas – Equipment cleaning and preparation – Airlocks – Gowning rooms – Hallways – Corridors – Control rooms
Group 2: Potential Contributors to Contamination			
Open, Functionally Closed, or Fully Closed System	• Product exposed directly to the environment • Open system connections	• Open system connections with local protection (i.e., LFH/UDF) • Functionally closed with sanitization (Risk level needs to be assessed based on the specifics of the set-up)	• Functionally or fully closed system • Functionally closed with sterilization (Risk level needs to be assessed based on the specifics of the set-up)
Preparation of Equipment/ Materials	• Cleaned equipment only • CIP of nondedicated equipment	• All equipment cleaned • Some parts sterilized • Some dedicated, some nondedicated equipment	• All equipment cleaned and sterilized • CIP/SIP; single-use consumables; all dedicated equipment
Complexity of Equipment Train/ Process	• Manual assembly or manual manipulations • High number of inputs into the system (manual or automated)	• Semi-automated with some manual manipulation • Medium number of inputs into the system	• Completely automated system • Low number of inputs into the system
Duration of Process Step	Multiple days of processing (>1 day)	4 hours–24 hours of processing	Less than 4 hours of processing
Process/Room Temperature	Warm (>25 °C)	Room temperature (20-25 °C)	Cold (2-8 °C)
Humidity Control	Absent	N/A	Present
Presence of Water, Media Powder Residue, or Other Growth-promoting Materials in Surrounding Area	• Open water source in area • Sink/drain present • CIP/SIP area • Tank draining • Safety shower/eye wash station present	Water/fluid/buffer/product transported to and used in area	• No transfer of fluids or solutions into area • Closed system piped directly to drain with air break

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(Table 4.3.1-2 Continued)

Criticality Factors	High Risk	Medium Risk	Low Risk
Group 2: Potential Contributors to Contamination			
Level of Activity in Surrounding Area (Personnel)	Common batch processing area (More than 1 operation occurring in room)	• Support area processing • Material/equipment prep area • Storage of sterilized parts and equipment	• Dedicated processing area • Storage of closed system vessels
Traffic Routes to/from Surrounding Areas	High traffic areas (e.g., hallways and airlocks)	Thoroughfares and access route to processing area (e.g., control rooms)	Low traffic areas or dedicated processing room (e.g., final formulation)
Group 3: Unusual/Unexpected Sources of Contamination			
Unusual/Unexpected Sources of Contamination	Unexpected contamination detected in an area, such as: • Leaking equipment/utility connection • Breach in integrity of environment • Damage to facility surfaces	N/A	N/A

4.3.2 Environmental Monitoring Risk Assessments

Risk assessments are used to provide rationale for EM locations and the frequency of sampling. The justification for site selection in each manufacturing room or area (ISO 5, ISO 7, and ISO 8) should be documented and included in the Environmental Monitoring Performance Qualification protocol for each area or in the change control for enhancement of the routine EM program.

For each major process step/segment/room in the manufacturing process, the items listed in **Table 4.3.2-1** may be assessed as risk control criteria to establish EM sample locations.

Table 4.3.2-1 Risk Criteria to Establish EM Sample Locations

Risk Rating	Sample Location Considerations
Low Risk	• Areas with no exposure to product pathway – Traffic patterns – Vectors of contamination (things people touch)
Medium Risk	• Process equipment exteriors or surfaces that are close to the open operations – Traffic patterns – Vectors of contamination
High Risk	• Each critical manufacturing process – Traffic patterns – Vectors of contamination – Process equipment exteriors or surfaces that are close to the open operations with exposure to product pathway

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4.3.3 Frequency of Monitoring

Based on the overall risk rating, the frequencies in **Table 4.3.3-1** may be considered. However, based on historical data, different frequencies may be more appropriate. If there is no historical data (e.g., for a new facility), a more frequent monitoring schedule should be utilized until data is obtained to support the development of an appropriate monitoring schedule.

Table 4.3.3-1 Frequency of Monitoring

Risk Rating	Frequency of Monitoring
Low Risk	Bi-weekly or weekly
Medium Risk	Weekly
High Risk	Each time a critical operation is performed

4.3.4 Conclusion

EM is an important element in bioburden control strategies for environments used for manufacturing processes that require bioburden control. Data derived from the environment and the process provide assurance that processes are being exposed to consistent and suitable environments commensurate with process requirements. Current guidance recommends using a risk-based approach to develop an EM program suitable to support manufacturing steps. The assessment of the critical factors identified in this technical report and presented methodology for executing a structured and robust risk assessment provide a scientifically sound approach to developing a risk-based EM program.

4.4 Risk Assessment Design: Example 1

This example summarizes results of a risk analysis conducted to evaluate the EM sample plans for a drug substance facility cell-culture area (not all details of the complete risk analysis process are included). A facilitator coordinated with participants from the manufacturing, quality assurance, validation engineering, and quality control groups and included a microbiology subject matter expert to conduct the risk analysis.

4.4.1 Executive Summary

This EM program is designed to monitor manufacturing areas and ensure that the levels of environmental controls are maintained in line with site SOPs. An EM risk analysis was conducted using a risk tool to identify and quantify risks in an effort to optimize the EM sample plans in the cell-culture area.

4.4.2 Discussion

A risk analysis team used a risk tool to evaluate contamination risks in all ISO-graded areas, choosing the EM sample site locations by the nature of operations to be carried out in the rooms. Sample site selection considered the various sanitized or sterilized surfaces, personnel, equipment, and material flow, as well as points of human activity that may represent potential risks to the state of control to the product and in the manufacturing area.

The risk analysis included the evaluation of each EM sample location for hazards. The hazards included the various physical conditions, operational activities, and EM-specific indicators that could contribute to or detect contamination. The identified hazards, factor descriptions, rationale, and weighting scores established for this analysis are provided in **Table 4.4.2-1**.

Each factor was assigned a weighting score of 0, 1, 3, or 5, based on its perceived contribution to contamination risk. Zero (0) equates to extremely low or no impact on contamination and 5 equates to the highest potential impact on contamination. The values were added to allow for a means to generate a risk priority.

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Table 4.4.2-1 Risk Assessment Design: Example 1 – Criticality Factors for ISO 7 and ISO 8 Environment Process Steps

Criticality Factors	High Risk (5)	Medium Risk (3)	Low Risk (1)
Group 1: Process Step			
Manufacturing Process Step	Critical manufacturing steps • Final formulation • Seed inoculation • Cell bank prep • Laminar flow/Unidirectional flow hood	Downstream operations • Cell culture • Purification • Sterilizer unload • Kitting of sterilized parts and materials	Upstream operations • Media/buffer preparation • Support areas – Equipment cleaning and prep – Airlocks – Gowning rooms – Hallways – Corridors – Control rooms
Group 2: Potential Contributors to Contamination			
Open, Functionally Closed, or Fully Closed System	Product exposed directly to the environment; open system connections	• Open system connections with local protection (i.e., LFH/UDF) • Functionally Closed with sanitization (Risk level needs to be assessed based on the specifics of the set-up)	• Functionally or Fully Closed system • Functionally Closed with sterilization (Risk level needs to be assessed based on the specifics of the set-up)
Preparation of Equipment/ Materials	• Cleaned equipment only • CIP of nondedicated equipment	All equipment cleaned; some parts sterilized, some dedicated equipment, some nondedicated	• All equipment cleaned and sterilized • CIP/SIP, single-use consumables, all dedicated equipment
Complexity of Equipment Train/Process	• Manual assembly or manual manipulations • High number of inputs into the system (manual or automated)	• Semi-automated with some manual manipulation • Medium number of inputs into the system	• Completely automated system • Low number of inputs into the system
Duration of Process Step	Multiple days of processing (>1 day)	4–24 hours of processing	< 4 hours of processing
Process/Room Temperature	Warm (>25 ºC)	Room temperature (20-25 ºC)	Cold (2-8 ºC)
Humidity Control in Room	Absent	N/A	Present
Presence of Water, Media Powder Residue, or Other Growth-Promoting Materials in Surrounding Area	• Open water source in area • Sink/drain present • CIP/SIP area • Tank draining • Safety shower/eye wash station present	Water/fluid/buffer/product transported to and used in area	• No transfer of fluids or solutions into area • Closed system piped directly to drain with air break
Level of Activity in Surrounding Area (Personnel)	Common batch-processing area (>1 operation occurring in room)	• Support area processing • Material/equipment prep area • Storage of sterilized parts and equipment	• Dedicated processing area • Storage of closed system vessels
Traffic Routes to and from Surrounding Areas	High traffic areas (e.g., hallways and airlocks)	Thoroughfares and access route to processing area (e.g., control rooms)	Low traffic areas or dedicated processing room (e.g., final formulation)

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The total hazard score for each of the EM sample locations was ranked and then filtered by the severity score for the room (**Table 4.4.2-2**). Severity is the measure of the possible consequences of a hazard potentially resulting in product contamination. Low severity was assigned if product contamination was unlikely as the product was never present in the room. Medium severity was assigned if the product was present but was never exposed to the room environment. High severity was assigned if the product was present and could be exposed to the environment.

Table 4.4.2-2 Risk Class Ranking Scheme

Hazard Probability	Severity		
	Low (1)	Medium (2)	High (3)
High	Risk Class 2	Risk Class 1	Risk Class 1
Medium	Risk Class 3	Risk Class 2	Risk Class 2
Low	Risk Class 3	Risk Class 3	Risk Class 3

The resulting risk class was then filtered by the detection controls (**Table 4.4.2-3**). Low detection is the use of the building monitoring system (e.g., HVAC, temperature monitoring systems) with no other means of detection. Medium detection includes the building monitoring system plus EM. High detection includes in-process product monitoring for contamination in addition to the building monitoring system and EM.

Table 4.4.2-3 Sampling Priority Designation

Risk Class	Low Detection	Medium Detection	High Detection
1	High sampling priority	High sampling priority	Medium sampling priority
2	High sampling priority	Medium sampling priority	Low sampling priority
3	Medium sampling priority	Low sampling priority	Low sampling priority

The output of the EM risk assessment is the level of risk at the EM sampling location expressed as a high, medium, or low EM sampling priority. The EM sampling priority provides information used to assess the EM sampling plan for each room by EM sample location. This ensures the identification of EM sample locations within the rooms that provide meaningful EM data and those sample locations that may be redundant or do not significantly contribute to the overall picture of environmental control. High EM sampling priority locations indicate room locations with the highest contamination risk or low detection capability.

These EM sampling locations were assessed to determine if additional remediation should be implemented to increase their environmental control. Medium EM sampling priority locations presented an acceptable level of risk and provided meaningful EM data. Low EM sampling priority locations were assigned in areas that either posed a very low risk of product contamination or sites that may have had higher risk classes, but the risk was mitigated by high-detection capabilities. The priority may have equated to a frequency or need to sample based on the potential to contribute to overall environmental contamination risk.

4.4.3 Conclusion

Warm Room X (**Figure 4.4.3-1**) is used to incubate the roller bottles. The room is maintained at 36 °C ± 1 °C. Activities in this room include introducing roller bottle racks, rotating roller bottles, removing roller bottles from the room, and inspecting daily for gross contamination or leaks of roller bottles; roller bottles will not be touched by humans. Due to the high traffic in the room and the higher than ambient temperature, sites in this room will be sampled weekly. Protecting the cleanliness of this room is key to the strategy of not contaminating roller bottles.

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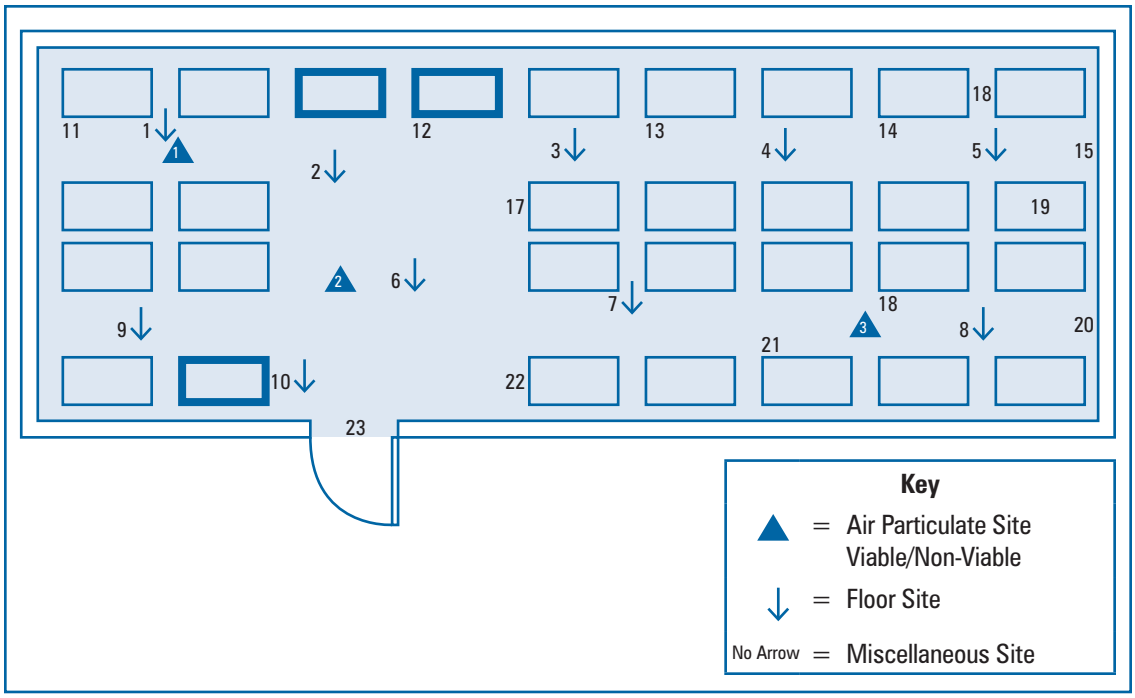
Figure 4.4.3-1 Warm Room X Layout and Monitoring Locations



The locations noted represent current routine sampling locations that are assessed annually to determine if they are appropriate, if there have been changes in the activities in the room, and if adverse bioburden events have occurred that may have resulted in targeted monitoring to determine root cause. **Table 4.4.3-1** provides details about the sampling site locations in Warm Room X.

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Table 4.4.3-1 Risk Assessment Design: Example 1 – Sample Site Location Details (Note: These are examples and do not describe all locations on the above diagram.)

EM Sample Site Locations	Temperature	Presence of Water, Other Growth-Promoting Materials	Processing	Duration of Process Activity	Sanitization Frequency	Processing Step	Room Activity	Preparation of Equipment / Materials Complexity of Equipment Train / Process	Grade/ Excursions	EM Sample Type Indicator	Overall Hazard Score
Site 1: Active air, warm room, near roller bottle incubation, north corridor	High (5)	Low (1)	High (5)	High (5)	Low (1)	Upstream (3)	Low (1)	High (5)	ISO 8 (3)	High (5)	34
Site 2: Active air, center of incubator	High (5)	Low (1)	High (5)	High (5)	Low (1)	Upstream (3)	Low (1)	High (5)	ISO 8 (3)	High (5)	34
Site 3: Active air, roller bottle incubation, south incubator corridor	High (5)	Low (1)	High (5)	High (5)	Low (1)	Upstream (3)	Low (1)	High (5)	ISO 8 (3)	High (5)	34
Site 1: Surface floor	High (5)	Low (1)	Low (1)	High (5)	Low (1)	Upstream (3)	Low (1)	Low (1)	ISO 8 (3)	Low (1)	22
Site 11: Surface rack-1	High (5)	Low (1)	High (5)	High (5)	Low (1)	Upstream (3)	Low (1)	High (5)	ISO 8 (3)	Medium (3)	36
Site 15: Surface wall-1	High (5)	Low (1)	Low (1)	High (5)	Low (1)	Upstream (3)	Low (1)	High (5)	ISO 8 (3)	Medium (3)	28
Site 12: Surface UDF	High (5)	Medium (3)	High (5)	High (5)	Low (1)	Upstream (3)	Low (1)	High (5)	ISO 8 (3)	Medium (3)	38
Site 13: Surface rack-2	High (5)	Low (1)	Medium (3)	High (5)	Low (1)	Upstream (3)	Low (1)	Low (1)	ISO 8 (3)	Medium (3)	26
Site 14: Surface wall-2	High (5)	Low (1)	Medium (3)	High (5)	Low (1)	Upstream (3)	Low (1)	Low (1)	ISO 8 (3)	Medium (3)	26
Site 21: Surface rack-3	High (5)	Low (1)	High (5)	High (5)	Low (1)	Upstream (3)	Low (1)	High (5)	ISO 8 (3)	Medium (3)	32
Site 9: Surface floor—south corridor	High (5)	Low (1)	Low (1)	High (5)	Low (1)	Upstream (3)	Low (1)	Low (1)	ISO 8 (3)	Low (1)	22
Site 10: Surface floor—near entry and exit door	High (5)	Low (1)	low (1)	High (5)	Low (1)	Upstream (3)	Low (1)	Low (1)	ISO 8 (3)	Medium (3)	24

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4.5 Risk Assessment Design: Example 2

This report summarizes results of the risk analysis conducted to evaluate the EM sample plan for a media supplementation (via aliquoting) process for a cell therapy manufacturing process.

Knowledgeable personnel from the microbiology, manufacturing, quality assurance, compliance, and manufacturing science and technology departments participated in the risk analysis.

In cell therapy manufacturing, media supplementation is performed by adding sterile supplements into sterile media using a sterile pipette or other sterile transfer method. The process occurs inside an ISO 5-classified UDF hood. Routine EM is performed based on the results of a risk assessment. Personnel are appropriately trained on the process and on the proper aseptic techniques. Personnel also wear sterile gloves and sterile sleeves while performing the process.

The media, supplements, and containers used during the process are sterile. In addition, an absence-of-contamination test is performed post-process using a rapid microbiology method.

4.5.1 Executive Summary

The EM program is designed to monitor manufacturing areas and ensure that the levels of environmental controls are maintained as specified in site SOPs and in the laboratory information management system. Using FMEA and following the risk assessment guidance methodology, an EM risk analysis was conducted to identify and quantify risks. The effort was intended to optimize the EM sample plans for media supplementation as well as to determine the appropriate ISO/Grade level needed. (See **Table 4.3.1-1** for recommended classifications and types of activities.) Other risk tool methodology may be used depending on the specific process.

4.5.2 Discussion

A risk tool was executed using appropriate methodology to evaluate contamination risks during media supplementation. EM sample site locations, including personnel monitoring, and the types of monitoring were determined based on the nature of the process. The nature of the materials (i.e., sterile), duration of the process, personnel, material flow, and post-processing testing were considered during the analysis. The risk analysis evaluated each EM sample location for hazards including the various physical conditions, operational activities, and EM-specific indicators that could contribute to or detect contamination (**Tables 4.5.2-1 and 4.5.2-2**).

Table 4.5.2-1 Risk Assessment Design: Example 2 – Criticality Factors for the Media Supplementation Process

Criticality Factors	High Risk (5)	Medium Risk (3)	Low Risk (1)
Group 1: Process Step			
Manufacturing Process Step **Media Supplementation**	• Addition of supplements to media via pipetting performed in UDF • Transfer of sterile media and supplements into the UDF (containers are closed, and outsides of the containers are sanitized prior to transport into the UDF) • Absence of contamination – rapid test method used at the end of process step		

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Table 4.5.2-2 Risk Assessment Design: Example 2 – Criticality Factors for Potential Contributors to Contamination

Criticality Factors	High Risk (5)	Medium Risk (3)	Low Risk (1)
Group 2: Potential Contributors to Contamination			
Open System • Pipetting Step	• Inadvertent touching of pipette or lid of containers via personnel gloved hand or sleeve • Leaning over open container, interruption of first air • EM performed		
Functionally Closed or Fully Closed System • Bottles during Incubation			• Leak in a sterile roller bottle • Caps not closed appropriately/leaking
Preparation of Equipment/ Materials			• Only sterile single-use equipment and materials used
Complexity of Equipment Train/Process	• Entire operation consists of manual pipetting (increased movement/handling of pipettes) • Under UDF, personnel gloves not sanitized properly		
Duration of Process Step • Pipette Additions		Aliquoting into each roller bottle, lids replaced on containers between each entry; additions performed using a fresh pipette for each roller bottle	
Process/Room Temperature	Process incubation at 37 °C		
Humidity Control in Room			• Present
Presence of Water, Media Powder Residue, or Other Growth-Promoting Materials in Surrounding Area			• No water used in the manufacturing process • WFI or purified water used in the preparation of disinfectants/ sanitizers • All "open" steps occur in UDF with no water present • No water drops in immediate manufacturing area; thus, no risk of puddles or other residual water in area
Level of Activity in Surrounding Area (Personnel)			• Minimal activities occur in surrounding area during the process
Traffic Routes To and From Surrounding Areas			• Minimal activities occur in surrounding area during the process; therefore, traffic minimized

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4.5.3 Conclusion

Based on the risk assessment, a routine EM was defined as seen in **Table 4.5.3-1**.

Table 4.5.3-1 Risk Assessment Design: Example 2 – Routine EM Program for Media Supplementation Process

Routine EM Program	
Personnel Monitoring	L-glove, R-glove, L- sleeve, R-sleeve at defined intervals in the media supplementation process (gloves and sleeves replaced if monitoring is performed prior to end of process)
UDF Hood Surfaces	Left, middle, and right surfaces; rear wall
Total Particulate Count	Prior to start of process
Active Air or Settle Plate	During process

Personnel monitoring was determined to be the best indicator in detecting contamination based on the criticality factors and would provide the most information on contamination sources in the process. The process relies upon aseptic technique and manual manipulation and transfer of media into containers, making personnel monitoring the best detection tool for the introduction of contamination. In addition, contamination is readily detected in the media bottles by turbidity and, typically, by a color change of the media from red to yellow, indicating acid production.

5.0 References

1. Parenteral Drug Association. *Technical Report No. 13 (Revised 2014): Fundamentals of an Environmental Monitoring Program*; PDA: Bethesda, Md., 2014.
2. European Commission. *Annex 1: Manufacture of Sterile Medicinal Products (corrected), EudraLex – Volume 4 – EU Guidelines to Good Manufacturing Practice,, Medicinal Products for Human and Veterinary Use* European Commission: Brussels, 2008(rev).
3. European Commission. *Guidelines on Good Manufacturing Practice Specific to Advanced Therapy Medicinal Products, EudraLex – Volume 4 – Good Manufacturing Practice for Medicinal Products for Human and Veterinary Use*; European Commission: Brussels, 2017.
4. Parenteral Drug Association. *Technical Report No. 70: Fundamentals of Cleaning and Disinfection Programs for Aseptic Manufacturing Facilities*; PDA: Bethesda, Md., 2015.
5. Pharmaceutical Inspection Convention/Cooperation Scheme (PIC/S). *GMP Annex 1 Revision 2008, Interpretation of Most Important Changes for the Manufacture of Sterile Medicinal Products*; PI 032-2. PIC/S: Geneva, 2010.
6. U.S. Pharmacopeial Convention. General Chapter <1115> Bioburden Control of Nonsterile Drug Substances and Products. In *USP 39–NF 34*, USP: Rockville, Md., 2016; p 1329.
7. European Agency for the Evaluation of Medicinal Products. *Decision Trees for the Selection of Sterilisation Methods: Annex to Note for Guidance on Development Pharmaceutics*; EMA: London, 2000.
8. U.S. Food and Drug Administration. *Guidance for Industry and FDA Staff: Current Good Manufacturing Practice Requirements for Combination Products, Final Guidance*; U.S. Department of Health and Human Services: Silver Spring, Md., 2017.

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9. Parenteral Drug Association. *Technical Report No. 29 (Revised 2012): Points to Consider for Cleaning Validation*; PDA: Bethesda, Md., 2012.

10. International Conference for Harmonisation. *Quality Guideline Q9: Quality Risk Management*; ICH: Geneva, 2005.

11. Parenteral Drug Association. *Technical Report No. 44: Quality Risk Management for Aseptic Processes*; Bethesda, Md., 2008.

12. International Conference for Harmonisation. *Quality Guideline Q10: Pharmaceutical Quality System*; ICH: Geneva, 2008.

13. International Organization for Standardization. *ISO 14644-1:2015 Cleanrooms and Associated Controlled Environments — Part 1: Classification of air cleanliness by particle concentration*; ISO: Geneva, 2015.

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Fundamentals of an Environmental Monitoring Program Annex 1: Environmental Monitoring of Facilities Manufacturing Low Bioburden Products

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